

boba_extension_dimensions

Boba Browser Extension — Research Dimension Decomposition

Dimension 01: Boba Project Architecture Deep Dive

Scope: Complete understanding of Boba-Base repo architecture, services, APIs, data models, plugin system, and how torrents flow through the system. **Key areas:** qBitTorrent-go service, FastAPI merge search, webui-bridge, download-proxy, plugin system, frontend dashboard, auth model.

Dimension 02: qBitTorrent WebUI API Complete Reference

Scope: Every endpoint, parameter, auth mechanism, and response format needed to add/manage torrents programmatically. **Key areas:** /api/v2/auth/login, /api/v2/torrents/add, /torrents/info, /sync/maindata, cookie auth, multipart upload, CSRF handling.

Dimension 03: Browser Extension Architecture (MV3)

Scope: Manifest V3 architecture patterns for torrent extraction extensions. **Key areas:** manifest.json, service workers, content scripts, popup pages, options pages, messaging (`chrome.runtime.sendMessage`), storage API.

Dimension 04: Cross-Browser Compatibility Matrix

Scope: Differences and compatibility shims for Chrome, Firefox, Opera, Yandex, Chromium. **Key areas:** Manifest V2 vs V3, `browser.*` vs `chrome.*` API, polyfills, store submission requirements, Yandex specifics.

Dimension 05: Tab Groups API Deep Dive (Chrome/Yandex)

Scope: How to query, enumerate, and extract URLs from tab groups in Yandex Browser and Chrome. **Key areas:** `chrome.tabGroups.query()`, `chrome.tabs.query({groupId})`, group permissions, extracting all URLs from a group.

Dimension 06: Magnet Link Detection & Parsing Algorithms

Scope: Comprehensive regex patterns, URI parsing, and infohash extraction from magnet links. **Key areas:** `magnet:?xt=urn:btih:` format, URI decoding, infohash validation (40-char hex), `dn/tr/xl` parameter extraction, edge cases.

Dimension 07: Torrent File Detection, Download & Bencode Parsing

Scope: How to detect .torrent links, download files, parse bencode, extract infohash. **Key areas:** .torrent link detection, fetch with CORS, bencode parser in JS/TS, SHA-1 infohash computation, File/Blob handling.

Dimension 08: Web Page DOM Scraping & Dynamic Content

Scope: Strategies for finding torrents and magnets on any web page including dynamically loaded content. **Key areas:** DOM traversal, MutationObserver for SPAs, common torrent site patterns, link detection, iframe handling.

Dimension 09: Extension ↔ Boba API Integration

Scope: How the browser extension communicates with Boba services and qBitTorrent WebUI. **Key areas:** REST API calls, CORS handling, authentication (API keys, cookies), WebSocket for progress, error handling, retry logic.

Dimension 10: Extension UI/UX Design Patterns

Scope: User interface components for torrent management extension. **Key areas:** Popup design, options page, context menus, notifications, badge counters, torrent list display, status indicators.

Dimension 11: Security Model & Privacy Architecture

Scope: Secure credential storage, permission model, CSP, and privacy considerations. **Key areas:** manifest permissions, host permissions, chrome.storage.local vs sync, secure key storage, content script isolation, CSP compliance.

Dimension 12: Testing, Build System & Store Distribution

Scope: Complete testing strategy and cross-browser build/packaging pipeline. **Key areas:** Jest for unit tests, Playwright for E2E, webpack/vite build, CI/CD, store submission packages (Chrome Web Store, AMO, Opera, Yandex).